

Ad Exposure Analysis

The Problem

Analyzing how repeated ad exposures influence CTR, conversion rate, and purchase likelihood to identify the exposure level that maximizes engagement while avoiding ad fatigue.

Key Questions

- How does exposure frequency affect CTR and conversion rate?
- What is the optimal exposure range before diminishing returns begin?

Which exposure level maximizes engagement?

Objective & Constraints

Objective

- Identify the relationship between ad exposure and engagement.
- Determine the optimal number of exposures before stagnation.
- Assess the effectiveness of different campaign types and channels.
- Provide data-driven recommendations for budget allocation.

Constraints

- No direct ad exposure count in raw data.
- No exact duration field for exposure frequency.
- Limited user-level behavior signals.
- Risk of ad fatigue needed trend-based validation.

The Data

Attributes

- Campaign type: Awareness, Retention, Conversion, Consideration
- Ad spend and exposure proxies
- Engagement: CTR, conversion rate, website visits, shares
- Demographics: age, gender, income
- Behavior: opens, clicks, prior purchases, loyalty points

Derived Variables

- Estimated ad exposure
- Viewer type: first-time vs. repeated
- Exposure duration buckets
- Campaign/channel performance groups

Operational Definitions

- Short-term: 1–3 days
- Medium-term: 1–2 weeks
- Long-term: 3+ weeks
- First-time viewers: lowest 10% of exposure proxy

Methodology

1. **Data Cleaning:** Standardized campaign types and handled missing values.
2. **Feature Engineering:** Built exposure, viewer-type, and duration metrics.
3. **Exploratory Analysis:** Used pivot tables and summary charts to inspect trends.
4. **Trend Analysis:** Measured how CTR and conversion rate changed with exposure.
5. **Recommendation Design:** Translated findings into budget and campaign guidance.

Analysis Logic

The study focuses on identifying the point at which additional ad exposure no longer produces meaningful gains, so that spend can be shifted toward higher-yield channels and campaign types.

Results & Implications

Key Findings

1. Best ad exposure range: **5–20 exposures** over **1–2 weeks**.
2. Exposure beyond **100+** shows diminishing returns.
3. **Consideration** campaigns delivered the strongest conversion rate.
4. **Retention** campaigns showed the strongest CTR.
5. **PPC** led CTR, while **Social Media** led conversion rate.
6. **\$1K–\$5K** spend range produced the best efficiency.

Implications for Advertisers

- Keep exposure concentrated in the early engagement window.
- Reduce over-retargeting and focus on acquisition.
- Allocate budget toward high-performing channels and campaign types.
- Monitor fatigue signals to avoid inefficient spend.

Conclusion

Ad performance improves up to a moderate exposure range, then starts to taper off as fatigue sets in.

The best strategy is to balance reach, timing, and channel selection instead of maximizing impressions alone.

Thank you

Ad Exposure Analysis Project